



Machine Learning Based Detection and a Novel EC-BRTT Algorithm Based Prevention of DoS Attacks in Wireless Sensor Networks

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Abstract

Wireless sensor network allows nodes to communicate with another using wireless channels so that it is necessary for packet delivery to the nodes. Black hole attack occurs when a malicious node receives and misleads a set of nodes in the wireless network to drop the data they received. Wormhole attack occurs when malicious nodes create data delivery latency between two nodes in the wireless network. In this paper, we propose a Machine Learning based Naive Bayes Classifier to detect and Enhanced Code-based Round Trip Time based method prevent those two vital attacks. The black hole attack can be analyzed using this technique with the help of authorize code. Similarly, the wormhole attack and fake destination attack can be prevented by trip time to find the data delay time to reach the destination. In the case of black hole attack, a node which does not have authorized code will be considered as malicious node and discarded. If the packet does not arrive at the destination within the specific time, it will automatically detect the occurrence of wormhole in the wireless network. The proposed method is experimentally validated on network simulator (NS2). The main advantage of the proposed method is to reduce the communication overhead.

Keywords Selective forwarding attack · Black hole attack · Wormhole attack · Channel based identification · Naive Bayes classifier · Enhanced code-based round trip time

1 Introduction

Wireless sensor networks (WSN) plays a very important role in various real-time applications starting from basic environmental monitoring, traffic management to very complex applications such as national defence and in other hard real-time systems. Basically WSN is a set of sensor unit which are interconnected through a wireless networks. These sensor networks collaborate to transmit data in a geological area. As these networks are very small in size they have many limitations such as storage, processing ability, and the aspects

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of energy also as they are normally provided with a very low power. There are many researches has been done to solve these problems to reduce the power consumption and to create energy efficient routing protocols such as LEACH [1], PEGASIS [2], TEEN [3] and APTEEN [4] etc. Through these techniques the energy efficiency of the routing protocols has improved very much, but the threat which is more important than the energy consumption is the wireless networks being vulnerable to various attacks such as Denial of Service attack, Black hole attack, wormhole attack, jamming attack etc. These attacks may lead to partial of completely disturb the network functionality and also it may consume needless power consumptions from the battery and reduces the lifetime of the sensor nodes. It is very important to protect the wireless network form these types of attacks.

In this paper, we propose a machine learning based Naive Bayes Classifier to detect and Enhanced Code-based Round Trip Time (EC-BRTT) based method prevent the DOS attacks. The black hole attack can be analyzed using this technique with the help of authorize code. Similarly, the wormhole attack and fake destination attack can be prevented by trip time to find the data delay time to reach the destination. In the case of black hole attack, a node which does not have authorized code will be considered as malicious node and discarded. If the packet does not arrive at the destination within the specific time, it will automatically detect the occurrence of wormhole in the wireless network. The main contribution of the proposed method is

- The proposed Enhanced Code-based Round trip time methodology is constructed to prevent black hole attack, wormhole attack and fake destination attacks.
- Selective forwarding attack detection is implemented to prevent the channel-based identification within the specified time frame.
- Naïve Bayes classifier a machine learning algorithm is used to identify the attacks using the behaviour of the node in Wireless Sensor Network.

The rest of the paper is organized as follows. The literature review is explained in Sect. 2. Enhanced Code-Based Round Trip Time is presented in Sect. 3. Experimental analysis is Sect. 4. And finally the conclusion and the future scope is discussed in the Sect. 5.

2 Literature Review

Networks and data can be easily affected by attacks leads to vulnerable without any security and control [5, 6]. Network must be kept safe and secure, and therefore security is quite important to secure the information from unauthorized users [7]. In Multi-hop wireless systems, security is more critical for transmission of important data between nodes [8]. In Wireless Sensor Network, any malicious node can join the network at any time [9, 10]. Malicious sensor nodes create loss of packets, change the routing path, add unnecessary messages and false characteristics [11]. Therefore, it is important to secure WSN from various attacks [12]. There are several types of attacks over the network such as the selective forwarding attack in which the attacker forwards each packet using Selective Forwarding Links [13]. Multi-hop networks are based on the assumption of forwarding the received messages to another node [14]. An unauthorized node accesses the packet transmission by routing and receiving the rebroadcast packets by attackers [15]. Malicious node performs as a normal node in Selective Forwarding Attack without

any proper security solution [16]. The malicious node will refuse to forward the messages and simply drop them. It ensures itself so that it cannot propagate the message further [17]. The intruders run the vulnerability that adjacent nodes will stop its operation and decide to seek another route [18]. Thus, a secure routing technique is required for finding the secure and optimum route between source and target destination [19].

There are several attempts in the literature to design secure routing. Hybrid DoS attack prevention technique was utilized to establish the trust-based energy consumption [20]. The prediction of energy utilization with power correlation checking procedure was evaluated to identify the detection rate in order to increase the lifetime of the network [20]. FFOA algorithm [21] was initialized to capture the attack node in the network. The compromised traffic is identified to reduce the energy cost. The localization for attack is computed using the Asynchronous Loss Tolerant Algorithm (ALTA) [22] to locate the attacks. Hidden Markov Model [23] was proposed to gather the optimization solution for capturing malicious nodes to prevent the black hole attacks. The optimality problem is rectified using the innovative methodology [24] that implements the Markov Decision Model to exploit the unknown attacks using cooperative spectrum. The efficiency of data communication in WSN was achieved by using the symmetric key based algorithm [25].

The ciphertext methodology is used to implement the encryption and decryption process in effective manners. The key size and the processing time are optimal to achieve the highest amount of data communication and reliability. The safeguard of the mobile sensor nodes in WSN from various kinds of security attacks is a critical task to implement. The Re-authentication strategy [26] was proposed to increase the energy efficiency by utilizing the mobile node authentication. The network traffic is reduced by implementing the methodology [27]. The load redistribution method was used to reduce the communication cost by identifying the attacks. The jamming attacks have been identified to increase the cyber security in the Medium Access Control mechanism using SIR [28]. This methodology is mainly used for layers protection in the network to provide the accuracy by identifying the specific attacks in the network. The level of security while providing the data communication in WSN has been improved by implementing the cryptographic methodology [29] that reduces the problems of using key management techniques. The group key generation will accurately implement the node security of the sensor nodes. The cluster-based group keys are used for implementing the simulation. The random number-based key generation is carried within the sensor nodes using the fuzzy membership function. The energy-related parameters of sensors have been limited with the condition of the battery. The energy utilization plays a vital role to design energy-based WSNs [30].

Hyper-Graph Clustering methodology [31] was designed to utilize the smaller amount of energy while data transmission. By this method, energy level parameters are well organized. WSN is embedded with Internet of Things (IoT) to improve smart services and computation. The edge computing technique was used to construct the distributed architecture for providing the speed response and improved quality-of-service (QoS). The vulnerability of attacks is the primary issue of data transmission. For high speed data transmission, the identification of monitoring sensor nodes is a challenging task in WSN. TF symmetric key method [32] was used to identify and avoid the attacks. The secured protocol was chosen for implementing the secured transmission in WSN. The authentication protocols within the single gateway node are widely implemented to provide secure and proficient authentication in WSN. The XOR based computation [33] was used to provide the security using session key within the users. Autonomous nodes were deployed in the WSN environment whenever the network size is very large.

The hybrid DFA methodology [34] has been proposed to provide the optimized secured route. The energy efficiency is achieved through the optimal path and also reduced the dead nodes in the network. The authentication based key exchange algorithm [35] has been proposed to improve the security with reduced vulnerabilities in WSN. The radio frequency related technologies are also used to increase the security parameters which are effective, secured and also more flexible in any condition. The new methodology SRPMA [36] has been proposed to provide the security with reduced energy consumption. The node trust-based evaluation methodology is utilized. An efficient security methodology [37] was proposed to overcome the security threats. The network lifetime will be affected with high amount of energy consumption in the network. The malicious nodes are identified to ensure the efficiency. The game theory approach was used to provide the optimal defence strategy in real-time systems [38–41]. Fast authentication process has been implemented to discover the DDoS attacks [42], AI based malicious attacks on Industrial IoT [43], the deep learning based Botnet detection system for IoT enabled smart cities [44] and LSTM model for providing security in smart grid [45] are the similar methods for providing security.

2.1 Limitations of the Existing Methods

- Systems will be compromised if an attacker deploys the combination of attacks, such as black hole, wormhole attacks.
- Does not define time synchronization that will leads the wormhole attack.
- Systems will not detect the fake destinations thus leads to black hole attacks.

3 Enhanced Code Based Round Trip Time Technique

The Enhanced Code-Based Round Trip Time (EC-BRTT) is an efficient method which is used to prevent wormhole and black hole attacks. This EC-BRTT method uses time bases scheme, including the time take to encode, transmission, received time, decode packets etc. Thus easily classify each packet with time difference. The Time Difference of Arrival (TDoA) technique will be used to detect the wormhole attack. The centralized system uses a cluster head called judge node that will record trust value of each node and the same will be available to nodes before transmission.

3.1 Advantages of EC-BRTT System

- Increase the Packet Delivery Ratio.
- Multi-hop routing and lower latency during transmission.
- Reduce the End-to-End Data Delay.
- Minimize the Communication Overhead.
- Improves the Attack Detection Accuracy.
- Updating the Judge Node Reputation Table while the attacks detected.
- Efficiently monitored the Packet Loss compared to other methods method.
- Deeply identified the false probability of EC-BRTT technique.

As shown in the Fig. 1 initially the Nodes are assigned and they are formed into regions according to channel based cluster using Wireless Sensor Network. Then it is moved into Selective Forwarding Attack Detection with the help of Judge Node selection using ETX

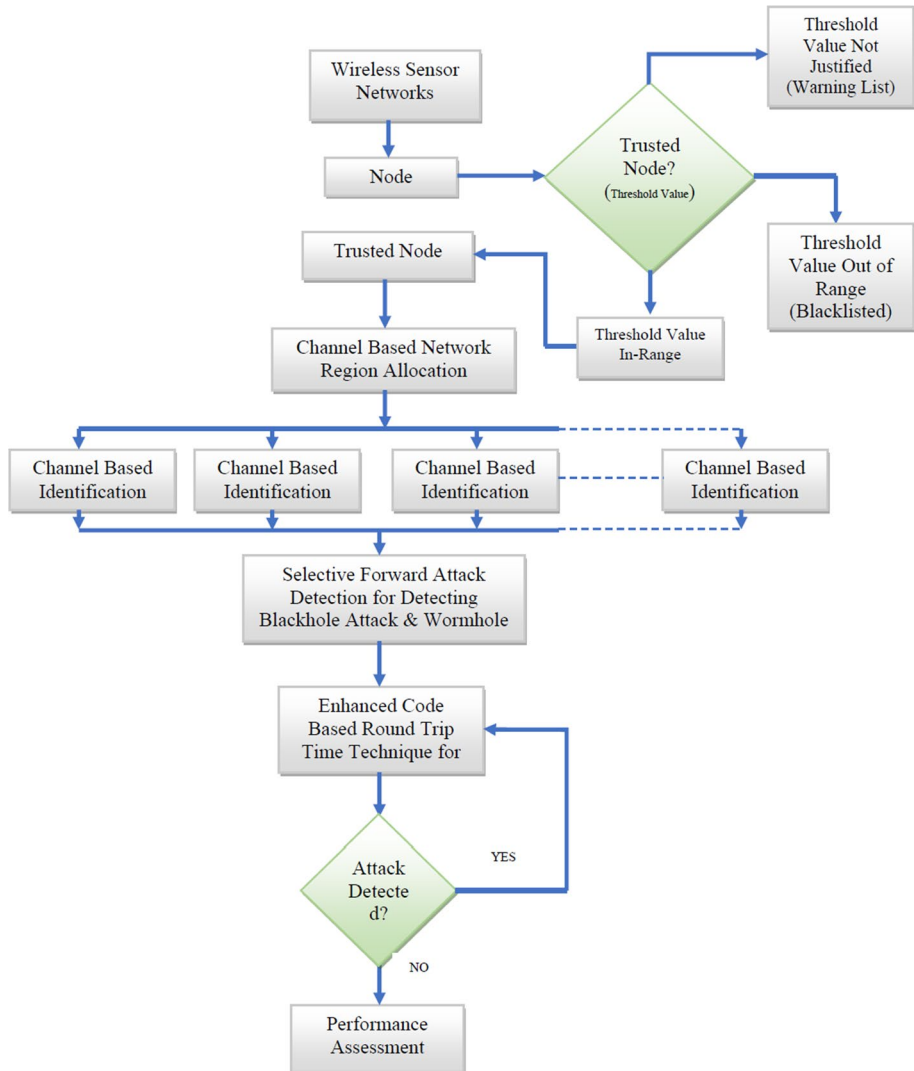


Fig. 1 Block diagram of the proposed system

threshold of each node in the cluster. This attack was detected by Channel Based Identification (CBI) method [46].

Now update the fields of routing packets by adding hop count, transmission path and estimated time elapsed. Here the proposed method considers Black Hole and Wormhole Attacks detection with time delay in selective forwarding attack. Proposed the Enhanced Code-Based Round Trip Time (EC-BRTT) technique to detect and prevent most two vital attacks with selective forwarding attack situation. Black Hole node is marked as Fake Destination and Wormhole node is marked as malicious node. The EC-BRTT method uses Naive Bayes algorithm to find the behaviour of the attack, whether it's an original attack or not. Both the attacks were detected and provide the result based on communication and attack. The performance can be evaluated in terms of Attack Detection, False Probability,

Packet Loss Monitor, Throughput, Packet Delivery Ratio, Data Delay and communication overhead.

3.2 Selective Forwarding Attack Detection

The selective forwarding attack detection is based on the reputation table which has grade points values of each node and detects whether the node is an attacker or not by using Naïve Bayes classifier and Adaptive Threshold algorithm. If the packet loss is more than identified packet loss, it is said to be an attacker [47]. Thus, the selective forwarding attack is detected by Channel Based Identification method composed of two operations namely Channel Identification and Data traffic Identification. This combination is used for detecting the selective forwarding attacks due to bad channel quality. This amalgamation-based detection monitors the combination of upstream and downstream which is configured by comparing its threshold levels.

3.3 Black Hole Attack Detection

This type of attack is a risky threat in wireless sensor network (WSN). If a node gets affected by this attack, it promotes itself as a shortest path and makes all other nodes to send the data towards itself [48]. In a wireless sensor network, if data are transferred from the source to destination, the source node finds its shortest path by sending route request packet to all its adjacent nodes. The adjacent nodes will in turn reply the source node with the route replay which consist of its sequence number and hop count. If one of the adjacent nodes is affected by the black hole attack, it will send a reply message with a highest sequence number and lowest hop number [49]. At this instant, the source node will receive reply from all the neighbouring nodes. In order to select the optimal path, the source node compares the reply from the entire neighbouring nodes with the sequence count and the hop count. As the node which is affected by the black hole attack has the highest sequence number and hops, it selects this node and forwards the data traffic to that particular node, and this node will absorb all the data and will not transfer the data to the destination [50]. Figure 2 demonstrates WSN under Black hole attack.

The proposed method is to discover the safe routes & avoid the black hole attack. By examining difference, the sequence number will be encapsulated to update packets and responses by source node [51]. For detecting black hole attack, an authorize code will be given to each node which assures the reliability of the node. Enhanced Code-based Round Trip Time method compare the authorization code with source node. If much more difference between source & destination exists, it is surely that intermediate node with differential authorization code is noted as malicious node and adds an entry to neighbour trust table [52]. All the nodes in the wireless network must have its own authorize code. A node that does not have authorized code will be considered as malicious node and will be marked in the topology and discarded [50]. Also, the nodes will verify the time difference between source & destination for each packet. The packets used to send information from the source to the destination contain ID of node who is sending this packet. Black hole attacker node will not send any data packet to cluster head. Judge node will not get any data from node which is a black hole attacker. Judge node waits for a fixed period of time [53]. If the node in the cluster doesn't send any packet even after this time period, it means an attacker exist in the cluster. The judge node remove attacker node from their routing table and call the procedure of clustering.

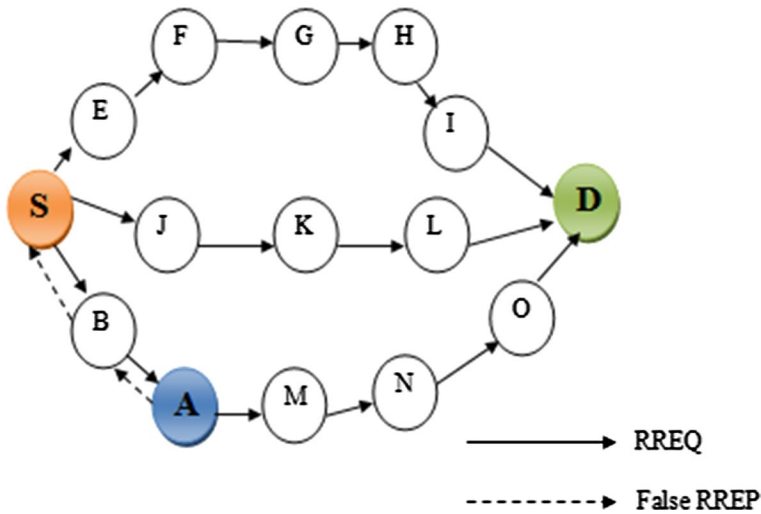


Fig. 2 WSN under Black hole attack

3.4 Wormhole Attack Detection

Wormhole attack is the next type of attacks which can cause very serious damages in the wireless sensor network. This wormhole attack requires two advertising nodes to implement the attack in the wireless sensor network. These two advertising nodes established a tunnel between them to transfer data. The wormhole attack is considered as the most vulnerable as it can initiate other dreadful attacks such as grey hole attack, sinkhole attack and even black hole attack. Figure 3 shows the wireless sensor networks under wormhole attack. Practically, nodes K and L are far from each other. In case of data transfer between two nodes, it requires more numbers of hops. As these nodes become an advertising node in wormhole attack, it forms a tunnel between two nodes K and L. The other nodes in the network conclude that the node K and L are neighbouring nodes and sends the data traffic to these nodes. Eventually, it disturbs the communication in the sensor network and drops the data packets. If in case node B needs to send a data packet to node B, it sends the RREQ to all its neighbouring nodes. The neighbouring nodes reply to the request with RREP. In this case, as node K and L are vulnerable, it also sends the RREP signal with a very low hop count. This actually requires more numbers of hop counts so that node B sends the data packets to node A and vice versa through this tunnel.

There are few techniques for detecting and avoiding Black hole attack and Wormhole attacks which operate on network layer. During wormhole deployment, the malicious route consumes higher time as compared to legitimate route. A malicious node captures the link on one end, transmits the packets through the wormhole link and replays them on another end. For detecting wormhole attack, each node will encapsulate the route request packet and send to the destination. The destination will reply the route response packet along with the nodes used for communication. Thus, the source node will gain the time taken to deliver the packet to the destination. If the packet does not arrive at the destination within the specific time, it will automatically detect the occurrence of wormhole in the wireless network. Also, EC-BRTT method can analyze the time taken with source and intermediate node for data generation, encoding, transmission, decoding, re-encoding & with destination

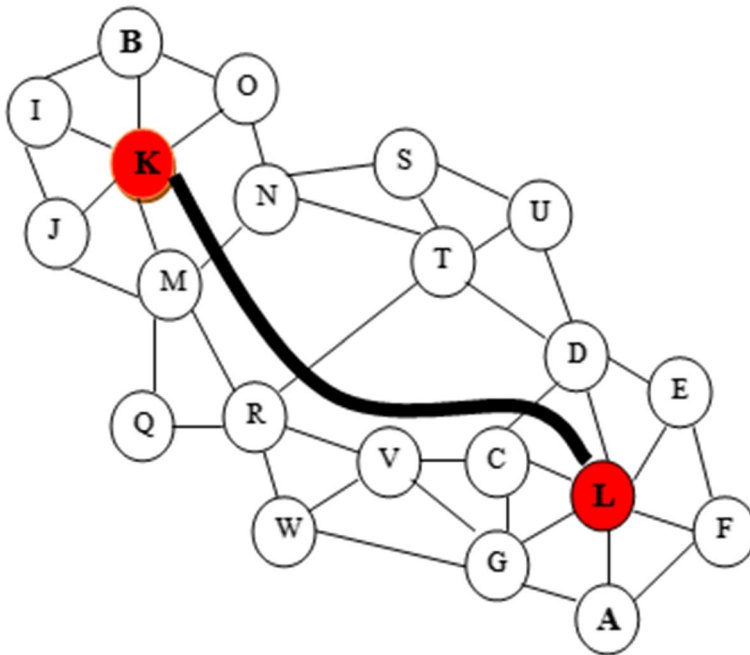


Fig. 3 WSN under Wormhole attack

for decoding, storing and error verification. Node with delay transmission will be identified and calculated the occurrence overall delay duration during all transmission. If the delay duration is higher than estimated, the intermediate node will be marked as malware/malicious node in its neighbour nodes. Here the proposed method also considers black hole and wormhole attack detection with time delay in selective forwarding attack. The proposed method uses Ad-hoc on demand Distance Vector (AODV) routing protocol which provides better results than optimized link state routing. Multi-hop routing is used. Round Trip Time method is used for finding time delay. This includes the route request, route response, route reply concept between source and destination. Round Trip Time (RTT) is also called as Round Trip Delay Time (RTD). RTT is the packet travel from specific source to a specific destination & back again which is illustrated in Fig. 4.

Generally, Round Trip Time (RTT) between two nodes can be calculated as:

$$RTT = Tr.D(R1) + Pr.D(R1 \rightarrow R2) + Q.D(R2) + Pro.D(R2) + Tr.D(R2) + Pr.D(R2 \rightarrow R1) \quad (1)$$

Tr.D → Transmission Delay from R1 & R2.

Pr.D → Propagation Delay from (R1 → R2) & (R2 → R1).

Q.D → Queuing Delay in R2.

Pro.D → Processing Delay in R2.

While approximating the round trip time the transmission delay, Queuing delay and processing delay can be made zero because it is considerably very low when compared with the propagation delay in case if the data transmitted is the first data and the routers does not

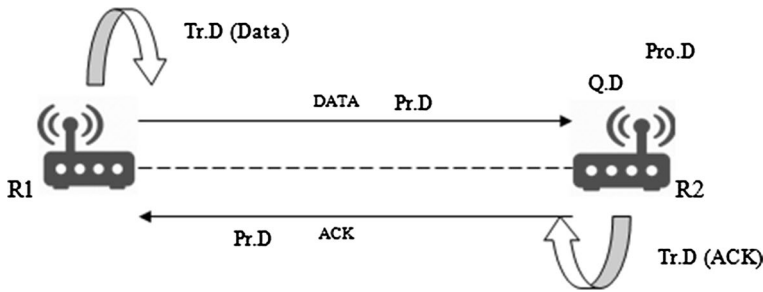


Fig. 4 Round trip time technique

have any previous data to transmit so the transmission delay $Tr.D$ and queuing delay $Q.D$ becomes zero. The processing delay is made zero because the data link layer just transfers the data packets to the network layer and data link layer gives the acknowledgement after considering all these nullifiers, Eq. (1) becomes

$$RTT \approx Pr.D(R1 \rightarrow R2) + Pr.D(R2 \rightarrow R1) \quad (2)$$

$$RTT \approx 2(Pr.D) \quad (3)$$

3.5 Enhanced Code-Based Round Trip Time

Step 1: Wireless sensor network is moved to nodes

Step 2: Channel is allocated for each node in the network

Step 3: Analyze the channel quality using Channel Based Identification

Step 4: Selective forwarding attack detection (Black Hole attack & Wormhole attack detection) using the results of the Channel based Identification method and detects the time delay in selective forwarding attack

Step 5: Application of Enhanced Code Based Round Trip Time Technique with the time delay in selective forwarding attack detection and detected Black hole and Wormhole attacks

Step 6: Implementation of Naive Bayes classifier algorithm to detect the attacks

Step 7: Performance analysis

The proposed Enhanced Code Based Round Trip Time technique describes that the Wireless Sensor Network is moved to nodes which were assigned according to channel based network region allocation. Here, the channel is allocated for each node which has a reputation value that tells which node can trust. It analyzes the channel quality (i.e., how much packet loss can occur). The grade points can be calculated by comparing identified packet loss and recently estimated packet loss. Depending upon the channel allocation and channel quality the grade points value can be fixed. From this, Data Traffic Identification is done to denote how many data packets send and loss. The proposed Enhanced Code-based Round Trip Time explains about flow of the Enhanced Code-Based Round Trip Time method. The Fig. 5 shows that nodes are initialized in WSN. Regions are formed based

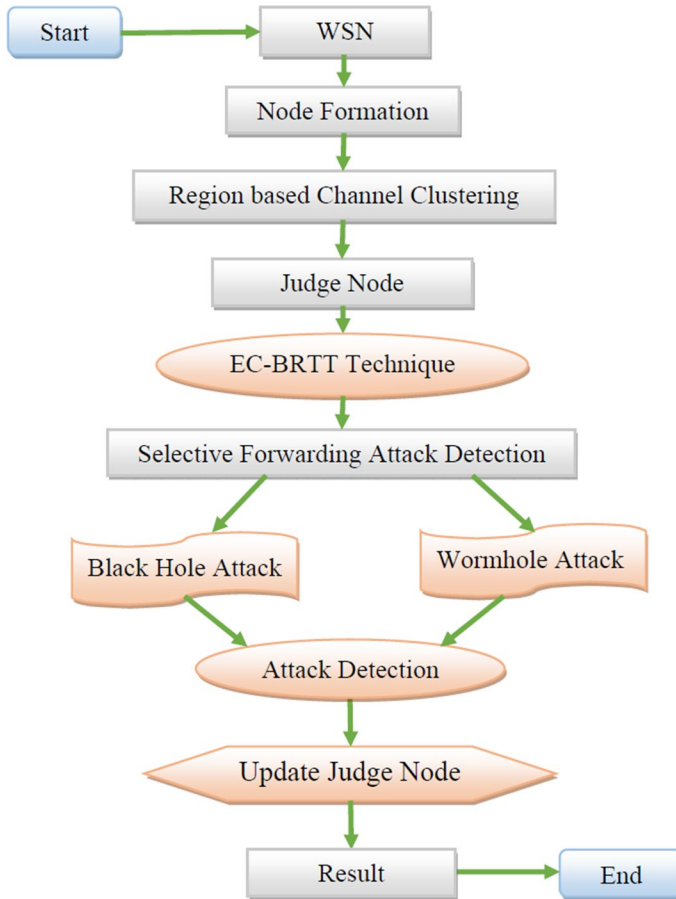


Fig. 5 The proposed enhanced code-based round trip time

upon channel clustering and set Judge Node for each region. Here analyze the black hole & wormhole attacks in SFA situation. Now detect the attack by using EC-BRTT Technique. Then update the Judge Node and finally provides Result.

Propose the Enhanced Code-based Round Trip Time technique is to detect and prevent most two vital attacks in selective forwarding attack situation. For detecting black hole attack, an authorize code will be given to each node which assures the reliability of the node. Enhanced Code-based Round Trip Time method compare the authorization code with source node. If much more difference between source & destination exists surely that Intermediate Node with differential authorization code noted as malicious node & adds an entry to neighbour trust table. All the nodes in the wireless n/w must have its own authorize code. If the node that does not have authorized code will be considered as Malicious Node & will be marked in the topology and discarded. Also the nodes will verify the time diff between source & destination for each packet. If a difference is much greater than average time then trigger the wormhole detection technique against intermediate nodes. During wormhole deployment the malicious route consumes higher time as compared to legitimate route. The Malicious Nodes captures the link on one end, transmits the packets through the

wormhole link and replays them on another end. For detecting wormhole attack, each node will encapsulate the route request packet and send to the destination. The destination will reply the route response packet along with the nodes used for communication. Thus the source node will gain the time taken to deliver the packet to the destination. If the packet did not arrive at the destination within the specific time, it will automatically detect the occurrence of wormhole in the wireless network. Also EC-BRTT method will analyze the time taken with Source and Intermediate Node for data generation, encoding, transmission, decoding, re-encoding & with destination for decoding, storing and error verification. Node with delay transmission will be identified & Calculate the occurrence overall delay duration during all transmission. If the delay duration is higher than estimated, then the intermediate node marked as malware/malicious node in its neighbour nodes.

Path Tracking is used to detect the behaviour of fake nodes through the route reply to the RREQ message. If a malicious node has received the RREQ, it will reply with a false RREP. When a malicious node replies with a false RREP by generating fake address list is recorded in the RREP. If node receives the RREP, it will separate the list by the destination address of the RREP in the IP field and get the address list. Then, node will determine the differences between the address lists recorded in the RREP. Each node in the same route which receives the RREP will do this same operation. Given that a fake node would reply the RREP to every RREQ, nodes that are present in a route before this action happened are assumed to be trusted.

The Enhance Code-Based Round Trip Time (EC-BRTT) used to find the time taken by a packet to travel from source to destination and back to source, which is directly proportional to the number of hops along the route. As number of hops increases, RTT of the corresponding route increases. In the presence of a wormhole attack, the number of hops will be very less compared to the actual route for the same source–destination pair, due to the presence of wormhole link. As the packets traverse a long distance, EC-BRTT of the route having wormhole link will be more compared to the normal route having same number of hops. Also EC-BRTT will update the judge node for each node with their trust value calculated against wormhole attack.

Response: Average Hop Count

$$\text{Avg. Hop Count} = \frac{\text{Sum (Hop Count of Transmission)}}{\text{Number of Transmission}} \quad (4)$$

Detection Ratio

$$\text{Detection Ratio} = \frac{\text{Sum(Total Detected)}}{\text{Sum (Actual Attack)}} \quad (5)$$

The Roundtrip time is computed in Eq. (6)

$$RTT = T_{Rep} - T_{Req} \quad (6)$$

where RTT, Round Trip Time; T_{Rep} , Time of Response; T_{Req} , Time of Request.

The attack detection is computed in Eq. (7)

$$\text{Attack Detection} = \frac{(Nodes - 1)\pi r^2}{RTT} \quad (7)$$

RTT Calculation for Intermediate Nodes is in Eq. (8)

$$RTT_{Node1,Node2} = RTT_{Node1} - RTT_{Node2} \quad (8)$$

where RTT_{Node1} , Round Trip Time for Node 1; RTT_{Node2} , Round Trip Time for Node 2.

For the node clustering process, the channel based clustering technique is used to identify the wormhole attack and find Malicious Nodes. Using hop count and timing analysis between the source and destination node, the presence of wormhole link along the route can be identified for verification. Once a route is marked for verification for wormhole communication, the judge node will process the wormhole attack detection technique against false alarm probabilities. For detection of wormhole attack, the fields of routing table should be modified to add a field called neighbour node. This will first confirm that the accurate transmission path. For adding the next node along the route, a verification process has to done to assure that the neighbour node is genuine. After cluster formation, two special route packets RREQ and RREP are used for confirming and localizing the presence of wormhole attack along the route. Figure 6 shows the timing diagram of the proposed methodology. Nodes are formed initially in WSN with Estimated Transmission Count (ETX) threshold. Formed regions based upon channel clustering. Select Judge Node in each region which has reputation table. Then starts communication between nodes and update JN Reputation table by calculating trust value with the help of hop count, transmission path and delay analysis. Verify communication whether the path is correct path or attacker path and update JN Reputation Table by finding attacks. Now detect both the black hole & wormhole attacks finally provide result in communication and attack parameters.

3.6 Naive Bayes Classifier

The Naive Bayes classifier algorithm is used to detect the behaviour of the node and other details like topology changes, cause of transmission intrusion, etc. Naive Bayes is used for calculating the posterior probability from Likelihood probability, class prior probability and predictor prior probability which is given by Eq. (9)

$$P(c|x) = \frac{P(x|c)P(c)}{P(x)} \quad (9)$$

$$P(c|X) = P(x_1|c) * P(x_2|c) * \dots * P(x_n|c) * P(c) \quad (10)$$

$P(c|x)$ denotes posterior probability of class (target) given predictor (attribute).

$P(c)$ denotes the class prior probability.

$P(x|c)$ denotes the likelihood probability of the predictor given class.

$P(x)$ denotes the predictor prior probability.

The Naive Bayes classifier is used in WSN because of its simplicity, elegance, and robustness. A large number of modifications have been introduced, by the statistical, data mining, machine learning, and pattern recognition communities, in an attempt to make it more flexible. Novel approach was proposed in to identify the attack node using Naïve Bayes classifier. The proposed Naïve Bayes framework was deployed for performing WSN attack node(s) detection. A new attribute, the end-to-end transmission time of each packet arrived at the sink is analyzed using Naïve Bayesian classifier for determining the network status. This technique doesn't involve any additional protocol and extra resource consumption of sensor node; it

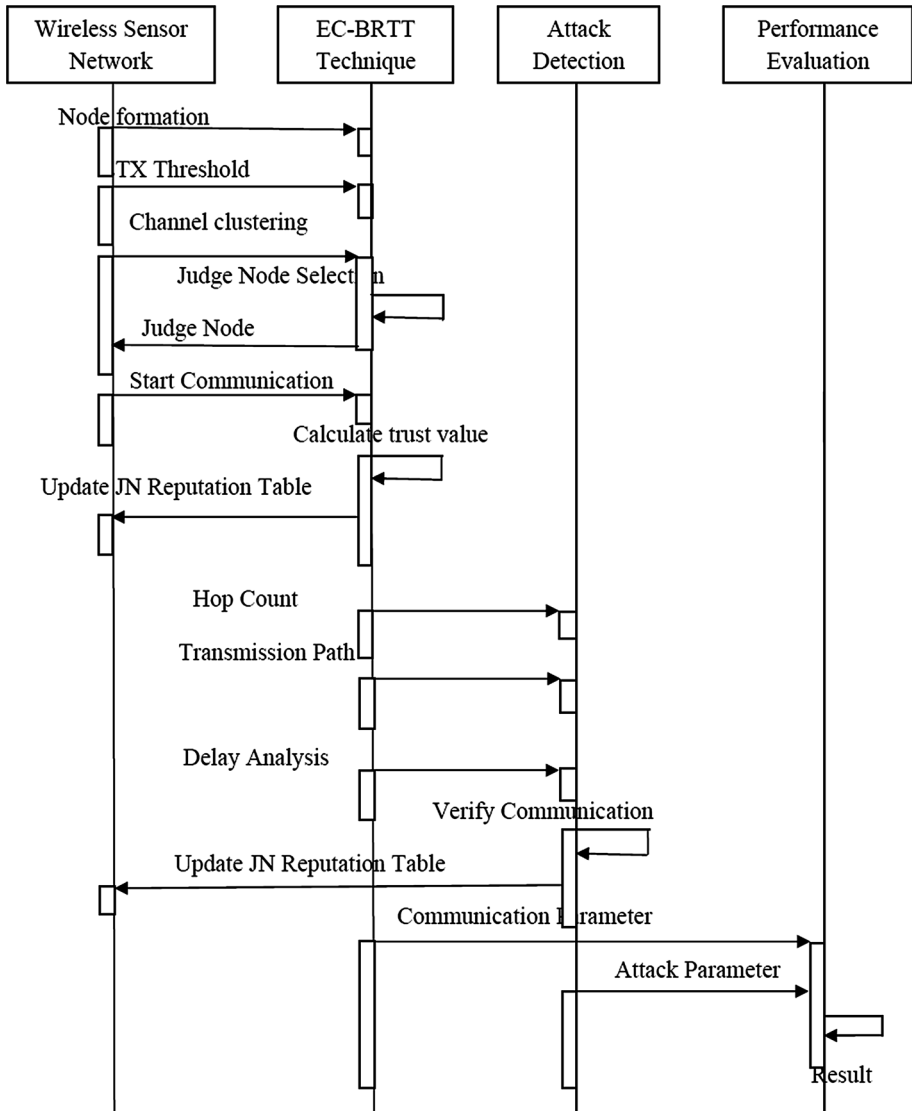


Fig. 6 Timing diagram

suggests a list of suspicious attack nodes to the user. Low computation complexity and High detection accuracy are the advantages of Naïve Bayes algorithm.

EC-BRTT PSEUDOCODE

start

Node Formation

for each node

Enable Channel Based Clustering

Select Judge Node based on ETX

Update the Routing Header

end

Initialize the Attack Detection Technique

Initialize Attack Phase

Initialize Enhanced Code-Based Round Trip Time

Get Number of Hops, Path, Latency Time

Create Trust Value to Judge Node

if (Number of Hops(route) = Actual Hops)

if (**not**(Path(route) = ActualRoute)

Mark node as attacker

end if

if (**not**(delay(route) = actual latency)

Mark node as attacker

end if

end if

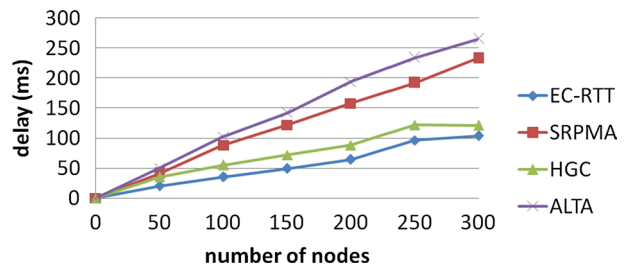
end

4 Experimental Analysis

The performance of routing protocol is measured with simulation results in NS2. Nam outputs and showed in graphs as Attack Detection, PDR, End-to-End Data Delay, Packets Loss Monitor, Throughput, False Probability and Communication Overhead. If the attacks are prevented communication overhead decreases and the PDR increases. The Black Hole Attack can be detected as Fake Destination using Authorizes Code. The Wormhole Attack can be detected as Malicious Node using hop count calculation, transmission path calculation and time elapsed calculation. The proposed method is compared with the related methods of ALTA [22], HGC [31] and SRPMA [36]. The parameter metrics used to implement the performance evaluation are Delay, packet delivery ratio, Energy consumption,

Table 1 Simulation parameters

Parameters	Values
No. of packets	8 packets
Average radio coverage range	175 m
Packet size	511 bytes
Packet transmission speed	45 bps
Channel data rate (threshold value)	11 Mbps
Packet dropping rate (threshold value)	18 bps
Average transmission delay in ms	7 ms

Fig. 7 Delay

Throughput, network lifetime, Detection rate and Residual energy. The simulation parameters are demonstrated in Table 1. The implementation is done in the following steps:

- Step 1: Initialize the Inputs.
- Step 2: Initialize the Parameters to be evaluated.
- Step 3: Initialize the nodes.
- Step 4: Identify the shortest path.
- Step 5: Identify the Malicious Nodes and Destination Nodes.
- Step 6: Identifying the fake Destination Nodes.
- Step 7: Performance Evaluation.

The Average hop count for the total amount of transmission is computed in Eq. (11)

$$\text{Avg. Hop Count} = \frac{\text{Sum (Hop Count of Transmission)}}{\text{Number of Transmission}} \quad (11)$$

Improved network lifetime and latency are the common parameters for scaling WSN. The nodes activate in a cycle to conserve energy, intermittently sense the environment and delivered the message to the sink node. If the data packet is not delivered within the specified time, the performance of the network is reduced and the delay is occurred in the network. Indeed, the network has more lifetime and preserve the energy can hold the minimized amount of delay. The proposed method compared to the related methods that the delay is minimized and it is demonstrated in Fig. 7.

To improve the lifetime of the network, the reduction of dead nodes is important. Whenever the dead nodes are minimum, the network lifetime is increased significantly. For reducing the dead nodes, the utilization of energy is very crucial. The proposed methodology improves performance to reduce the number of dead nodes in the network and increases

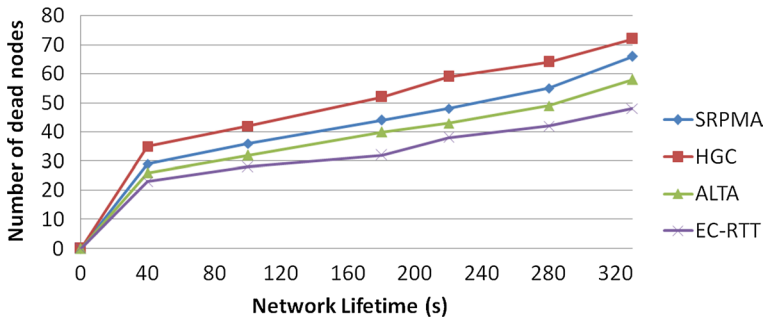


Fig. 8 Number of dead nodes with network lifetime

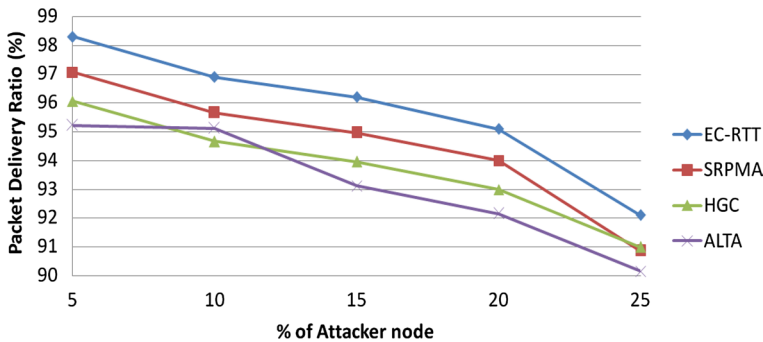


Fig. 9 Packet delivery ratio

the network lifetime. Figure 8 demonstrates the network lifetime compared with number of dead nodes. Figure 9 demonstrates the packet delivery ratio which is compared with the related methods. Whenever the attackers are increased significantly, the packet delivery ratio will be reduced in reasonable amount. Prevention of the DoS attack is the permanent solution to increase the packet delivery ratio. The proposed method has improved identification of the DoS attacks compared to the other related methods.

The energy utilization depends on the communication from the sender to the receiver in spite of distance parameter. If the network has minimum amount of active nodes, the performance of the network is reduced. To increase the active nodes in the network, the dead nodes should be minimized. The proposed methodology has increased number of active nodes, so that the energy consumption is reduced. It is performed well compared to the other related methods. It is demonstrated in Fig. 10.

Throughput is measured using the total number of data packets delivered to the receiver using the sink node in the network. It is commonly called as the bandwidth consumption. The queuing theory formation is applied to compute the throughput. Whenever the dropped packets are reduced, the throughput is increased. Throughput may be affected in several factors like network lifetime, residual energy and energy consumption. The simulation results proved that the proposed methodology has increased amount of throughput compared to the related methods demonstrated in Fig. 11.

The detection rate provides the accuracy for detecting the DoS attack in the network to increase performance. The detection rate is computed in Eq. (12)

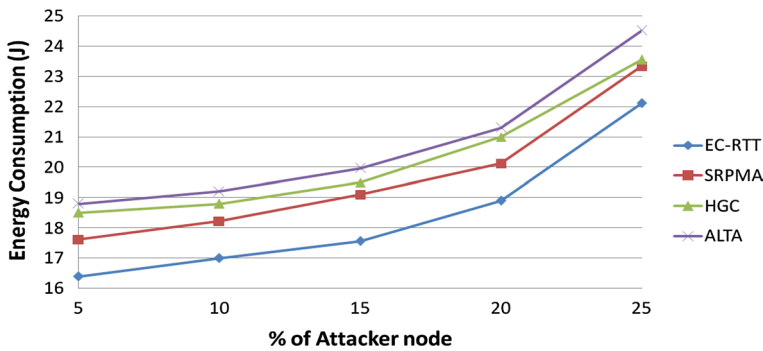


Fig. 10 Energy consumption

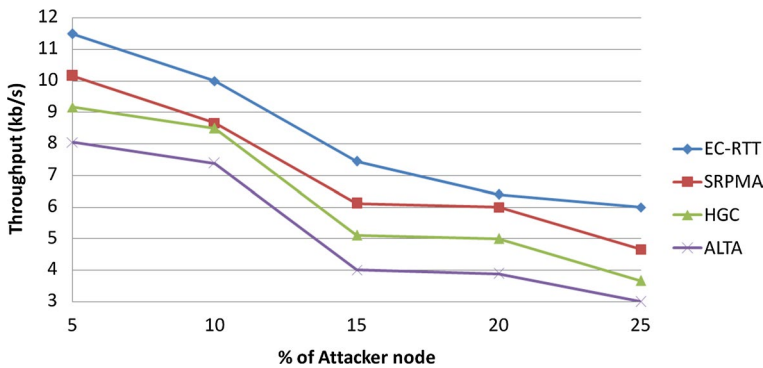


Fig. 11 Throughput

$$\text{Detection Ratio} = \frac{\text{Sum}(\text{Total Detected})}{\text{Sum}(\text{Actual Attack})} \quad (12)$$

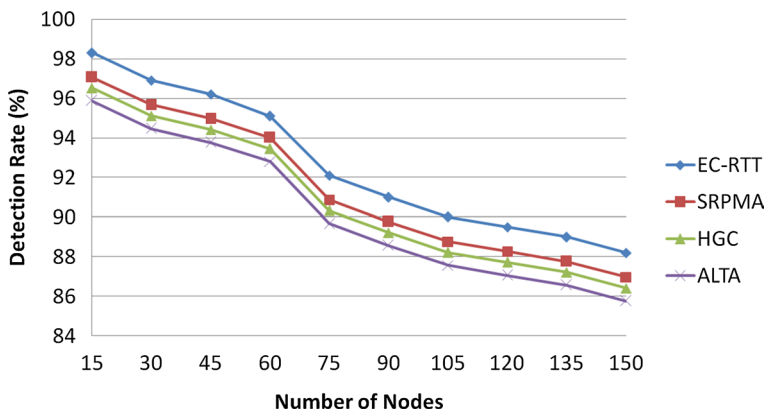
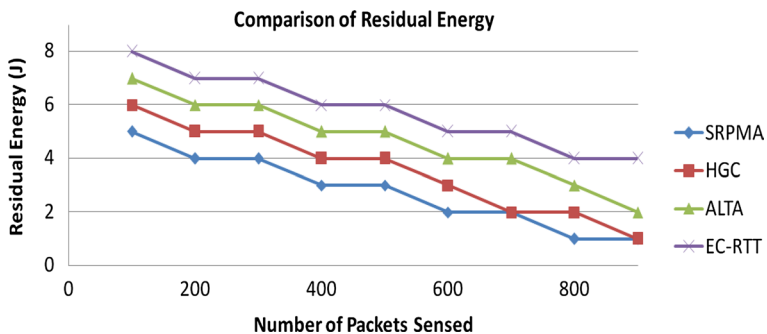
Table 2 demonstrates the detection ratio for the proposed technique with the related techniques.

If the normal node is asserted as the attacker node then the drop packets are increased. The simulation is performed in different times and the results are analyzed. The results proved that the proposed work has improved detection rate compared to the related methods demonstrated in Fig. 12.

The energy model demonstrates the level of energy for the nodes in WSN. The node has an initial level of energy at the simulation in a specified amount of time. A node drops an amount of energy to transmit the data packet and received the data packet. Whenever the node involved in the active transmission, energy is reduced. The energy level of the particular time of involving the data transmission is calculated using the residual energy. Network traffic plays a vital role for computing the residual energy within a specified time period. The simulation is performed in regular interval to find the residual energy of the particular node. If more number of packets is sensed means the residual energy level is reduced. The simulation results prove that the proposed methodology has increased amount of residual energy compared to the related methods demonstrated in Fig. 13.

Table 2 Detection ratio

Number of nodes	EC-RTT	SRPMA	HGC	ALTA
10	0.97	0.95	0.93	0.92
20	0.96	0.95	0.93	0.92
30	0.95	0.94	0.92	0.91
40	0.95	0.94	0.92	0.91
50	0.94	0.93	0.92	0.91
60	0.94	0.93	0.91	0.90
70	0.93	0.92	0.91	0.90
80	0.92	0.91	0.90	0.89
90	0.92	0.91	0.89	0.88
100	0.91	0.90	0.88	0.87

**Fig. 12** Detection rate**Fig. 13** Residual energy

Selective forwarding attack is the common routing attacks in WSN, since the attacker node drops the data packets to decrease the network performance. The sink node forwards the data packets through the communication link with transmission range. The attacker

node tries to become a sink node to forward the data packets. The proposed scheme identifies the DoS attacks and prevents the attack in WSN. The participant node which has the reduced amount of battery level for accurate sensing is called as the dead nodes, these nodes can be reduced by identifying the attacks and maintaining the effective energy utilization.

Figure 14 explains that in the Green colour region, the shortest route is 17-10-51. The 16th node as Malicious Node (Red), 14th node as Destination (Blue) & 12th node as Fake Destination (Orange). Then in the Purple colour region, the shortest route is 32-24-0-9. The 8th node as Destination (Blue), 50th node as Malicious Node (Red) & 7th node as Fake Destination (Orange). Thus the final output shows that the shortest path and found Malicious Node (Wormhole Attack), Fake Destination (Black Hole Attack), Destination for Pink, Yellow, Green and Purple colours regions detection.

5 Conclusion

In this paper, we proposed a secure routing technique or finding the secure and optimum route between source and target destination. A Novel Naive Bayes machine learning classifier was developed to detect and Enhanced Code-based Round Trip Time (EC-BRTT) technique was developed to prevent the most two vital attacks in selective forwarding attack situation. This method finds out the time of packet and forwards it to the judge node so that judge node will easily find out the average time taken to the destination. This technique is centralized which also avoid communication overhead. The algorithm used to detect attacks is Naive Bayes classifier. An EC-BRTT uses trip time to find the data delay time to reach the destination in order to find the wormhole attack (malicious node) and black hole (fake destination) attack. The simulation results showed that malicious node and fake destination node from the set of nodes. The output graphs are very effective in terms of Communication Overhead, Data Delay and Attack Detection. In the future scope, the proposed

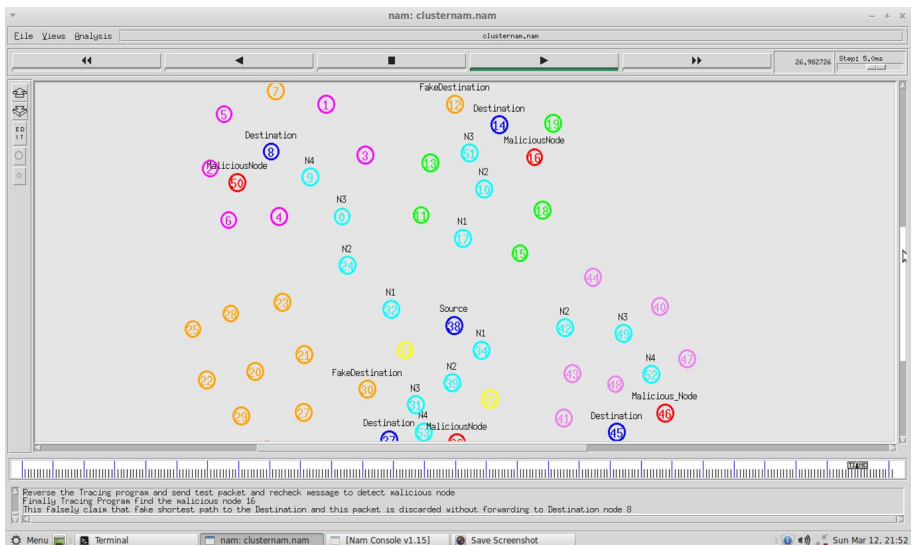


Fig. 14 Final output detection. (Color figure online)

Table 3 Abbreviation and symbols with meaning

Abbreviations	Acronyms
WSN	Wireless sensor networks
EC-BRTT	Enhanced code-based round trip time
NS2	Network simulator 2
DoS	Denial of service
FFOA	Fruit fly optimization algorithm
ALTA	Asynchronous loss tolerant algorithm
SIR	Signal-to-interference ratio
IoT	Internet of things
QoS	Quality-of-service
DFA	Deterministic finite automata
SRPMA	Secure routing protocol based on multi-objective ant-colony-optimization
DDoS	Distributed denial of service
LSTM	Long short term memory
TDoA	Time difference of arrival
RREQ	Route request
RREP	Route reply
AODV	Ad-hoc on demand distance vector
RTT	Round trip time
RTD	Round trip delay time
Tr.D	Transmission delay
Q.D	Queuing delay
SFA	Selective forwarding attack
ETX	Estimated transmission count
JN	Judge node
PDR	Packet delivery ratio
HGC	Hyper graph based clustering

technique will be injected in the routing protocols so that it will provide more transmission. Also, the proposed model can be enhanced by providing encryption algorithm that satisfies both integrity and authentication (Table 3).

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